



Advancements in natural remediation management techniques for oil spills: Challenges, innovations, and future directions

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ABSTRACT

The natural remediation of oil spills offers a promising avenue for addressing environmental disasters, leveraging the inherent power of biological, chemical, and physical processes to restore affected ecosystems. Techniques such as bioremediation, phytoremediation, and natural attenuation hold significant potential for mitigating the devastating impacts of oil spills while minimizing environmental harm. However, several challenges must be overcome to optimize the effectiveness of natural remediation methods for oil spills. These challenges include the variability in oil composition, environmental conditions, and the availability of suitable microorganisms or plants for degradation. Additionally, the scalability of these techniques for large-scale spills, concerns about unintended ecological consequences, and public acceptance pose significant hurdles. Looking toward the future, ongoing research and innovation offer promising directions for advancing natural remediation methods for oil spills. Biotechnological advancements, including developing genetically engineered microorganisms with enhanced oil-degrading capabilities, hold promise for accelerating remediation processes. Integrated approaches that combine natural remediation techniques with traditional cleanup methods may offer synergistic benefits, maximizing the effectiveness of oil spill response efforts. Furthermore, enhancing monitoring and predictive modeling capabilities can improve our ability to assess the effectiveness of natural remediation efforts and predict their long-term environmental impacts. Standardizing protocols and guidelines for implementing natural remediation techniques in response to oil spills can facilitate regulatory approval and streamline response efforts.

1. Introduction

All nations compete intensely for petroleum oil, a vital economic commodity [138]. The petrochemical sector thrives because human activity depends on oil to supply its energy needs although excessive using of petroleum deteriorates our ecosystem [221]. Petroleum-based hydrocarbon spills and discharges are common throughout the lifecycle of petroleum products, including during manufacturing, storage, refining, transportation, and processing. Incidents can occur due to oil-field blowouts, leaks from pipelines and storage tanks, accidents involving oil tankers, oil well waxing, and malfunctions in refinery and petroleum manufacturing equipment [215]. Big spills should be recycled or cleaned up as much as possible. Still, occasionally it is

impossible to get the materials back, so they stay in the impacted region and continue to pose environmental problems. Therefore, whenever oil is extracted, there is always a risk of pollution when combined with a lack of capacity to handle settings tainted by oil, particularly in harsh or unusual places like polar regions, deserts, deep sea areas, and marshes. The ecology that produces oil has had a terrible time managing this catastrophe [141]. Severe and long-term impacts on wildlife and plants may result from the presence of hazardous substances such as hydrocarbons, heavy metals, sulphur compounds, nitrogen-oxygen compounds, and spilled crude oil or refined petroleum products like gasoline or lubricant oils [157]. The cleanup is complicated due to the toxicity and complexity of hydrocarbons found in crude oil [188]. Known techniques for controlling this contaminated site include

Abbreviations: PAHs, Polycyclic aromatic hydrocarbon; TPH, Total petroleum hydrocarbons; PH, Petroleum hydrocarbons; BS, Bio-stimulation; BA, Bio-augmentation; AOP, Advanced oxidation process; SOM, Soil organic matter

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biological, thermal, physical, and chemical processes, which can be applied singly or in combo depending on the pollutant's type and the amount and the environment [191]. While each approach has advantages and disadvantages, both chemical and physical methods which are regarded as major methods are employed for expedient remediation and to stop oil from accumulating [190]. The usual projected cost of cleaning up the crude oil leak has been \$2730 per barrel of oil [69]. Compared to breaches that have happened closer to coastal areas and populated areas, those that occur in open waters distant from beaches and shorelines have a significantly less financial effect on the ecology and need far less money for cleaning. According to reports, the anticipated cost of cleaning up the shoreline is 100 times more than the cost of pumping from a damaged vessel and four to five times higher than the expenditures incurred in cleaning up an oil leak in wide waters [56]. The cost of cleaning likewise rises as the amount and rate of spills grow. For every percentage rise in spill size, the cost of losses rises by almost \$718,000 [15].

The removal of oil from water can be accomplished using a variety of oil remediation methods. Biological remediation, Physical remediation, thermal remediation, and chemical remediation are the four main categories into which the currently available treatments may be divided. Strategies for physical cleanup include skimmers, sorbent materials, and booms. Oil booms contain oil spills and stop them from expanding. The spills are then cleaned up with the use of skimmers, sorbent materials, or other remediation techniques. Dispersants and solidifiers are used in chemical remediation processes to clean up oil spills [68]. Microbes are used in biological remediation procedures to accelerate the natural breakdown of oil constituents [228]. Waves are used to contain oil spills, and thermal cleanup procedures include merely lighting the booms. The best strategy will rely on several variables directly or indirectly impacting the repair procedures. A few of these variables include the ambient temperature and tidal wave patterns at the spill site, the volume of the oil spill, the characteristics of the oil spill, and the threat the oil spill poses to the surrounding environment, the human population, important sites for commerce and culture, and [58]. This pollution from the leak of oil needs to be cleaned up immediately. One way to do this is by using phytoremediation, a crucial component of biological remediation that focuses only on using plants to absorb, contain, or remove pollutants from their surroundings through absorption or transformation [160]. It has been demonstrated that several plant species are capable of growing in polluted soils and removing pollutants from the growth media; these plants may accomplish this on their own or in cooperation with microbes [160]. Because there are naturally occurring petroleum hydrocarbons (PH) in the environment, which can be broken down by microorganisms have developed as a consequence. These kinds of microorganisms might be used to cure oil [228]. As a result, waste materials from the culinary, agricultural, chemical-based, and medicinal sectors have been tested for and used with microorganisms. Due to its relatively low expense and environmentally benign nature, using bacteria to treat environmental toxins has become a promising technique in the past few years [101]. An innovative technique for cleaning up petroleum contamination has also been made possible by the ongoing advancement and growth of microbial remediation technology, and it has garnered a lot of interest [73,77].

Therefore, it is usually preferable to either stop oil spills wherever feasible or take quick action to lessen their harmful effects. Nonetheless, unintentional oil spills might still occur, and in those cases, it's critical to use all possible oil spill cleanup methods.

Addressing the gaps in oil spill remediation can contribute to the development of future response and repair solutions that are more ecologically friendly, equitable with society, and efficient. The effectiveness of novel remediation techniques is evaluated by contrasting new technologies, such as bioremediation or nanotechnology, with more established techniques, such as booms or dispersants, in a variety of environmental settings and oil spill scenarios [190]. Improved risk assessment models and mitigation techniques are needed to anticipate and lessen any unforeseen ecological effects or public health hazards

that may arise from remediation activities [58]. This study aims to assess the efficacy of current rules and regulations about oil spill response and remediation. The research will utilize case studies and comparative analysis to find areas for improvement [35]. Combining mechanical, chemical, and biological cleanup techniques into integrated systems to maximize efficacy and reduce environmental effects during oil spill disasters is being investigated [190].

This review examines the effectiveness of diverse oil spill cleanup methods: biological, physical, thermal, and chemical approaches. Each method has specific strengths and limitations in addressing environmental and economic impacts post-spill. Environmental repercussions on ecosystems, biodiversity, and water quality are critical, as are the financial impacts on industries and communities. Urgent remediation is emphasized to minimize long-term ecological and socioeconomic damage. Factors influencing cleanup strategy selection include spill location, volume, oil type, and environmental sensitivity. Tailoring methods to these variables optimizes effectiveness and minimizes collateral damage. Recent advancements in cleanup technologies are integrated, assessing innovations in containment, recovery, and treatment. Emerging trends include the integration of multiple strategies for enhanced efficacy across different environmental settings. Future challenges include improving response times and effectiveness, fostering technological innovation, and enhancing international collaboration and policy frameworks. Ultimately, the review aims to advance environmental stewardship and sustainable practices in oil spill response, addressing current gaps in knowledge and proposing strategies for mitigating future incidents.

2. Material and methods

2.1. Conducting a literature review and gathering data

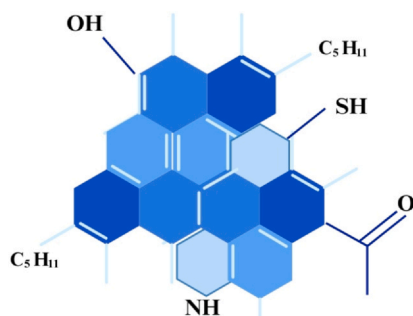
Articles published up until April 2024 were included in the literature search, with a focus on examining the harmful impacts of oil on the environment and possible corrective measures. The search used terms including "oil," "oil spills," and "oil spill remediation," utilizing international publications and online databases such as Web of Science, Science Direct, and Scopus. Predetermined criteria were set to only include publications that contained quantitative data on the toxicity of oil spills to guarantee objectivity. For every result that was found, a methodical, multi-step evaluation was part of the screening process:

1. Title and Abstract: To ensure that the title and abstract were pertinent to the discussion of oil spill toxicity and/or possible management options, a preliminary review of them was done.
2. Abstract: To determine if the abstracts of the chosen articles from Step 1 were in line with the goals of the review, a thorough analysis of them was conducted.
3. Full Content: Every article that was screened in Step 2 was examined in its entirety.
4. References: Every document mentioned in the articles screened in Step 2's "References" section was thoroughly examined.

Following these stages of screening, 132 articles were chosen for additional review.

3. The consequences of spilling crude oil

Natural compounds called hydrocarbons found in crude oil are derived from marine vegetation that has been deposited over thousands of years [132]. They are primarily made up of substances that vary in dissolution, fluctuations, and biological degradation susceptibility. The asphaltic fraction (Fig. 1) with double covalent bonds and aromatics with more condensed rings, which are more resistant to biodegradation, are found in spilled oil together with aliphatic (like alkanes and alkenes) and aromatic molecules that are highly susceptible to destruction [8]. Identifying the oil hydrocarbons and forecasting both the immediate and long-term tendencies of spilled oil are essential for an oil



The structural features of asphaltenes, which stabilize oil-water emulsions, include aliphatic chains, aromatic rings, and heteroatoms (S, N, and O) in functional groups.

Fig. 1. The structural features of asphaltenes, which stabilize oil-water emulsions, include aliphatic chains, aromatic rings, and heteroatoms (S, N, and O) in functional groups.

spill's effective cleanup. Several minor spills happen on waterways and wetlands every year, causing both acute and long-term harmful effects on wildlife and plants and occasional big spills that significantly affect coastlines and wetlands [201]. A petroleum spill's physical and hazardous characteristics are impacted by several modifications in composition it encounters as it reaches the aquatic environment [228]. Most of the unstable and hazardous elements evaporate, and just a small portion of them become oxidized by sunlight's UV rays [132]. Certain low-molecular-weight hazardous chemicals dissolve in bodily fluids and undergo rapid degradation [30]. Some of them attach themselves to small particles in the water and sink to the bottom [35]. Large amounts of the oil are broken down into tiny droplets, scattered across the water's surface, and left there to be destroyed by microorganisms [225]. Occasionally, the droplets combine to create a mousse, or emulsified of water and oil, which lengthens the slick's tenacity [127]. Petroleum leaks may take place on terrestrial in addition to in marine environments. The destiny of PH and the degree of plant damage are determined by the kind of soil (sand, clay, and loam,) and the degree of living things present [190]. Monocyclic aromatic hydrocarbons, which have a low molecular weight and high solubility, are among the constituents of crude oil that often vaporize into the environment after spilling on soil. Native bacteria often break down the C10–C16 n-alkanes. Further molecular weight fractions, on the other hand, are more susceptible to biodegradation. Examples of these include alkanes with numbers of carbons more than twenty, and polycyclic aromatic hydrocarbons (PAHs) including anthracene, naphthalene, pyrene, phenanthrene, and their alkylated analogues. Because of these heavy chemicals' capacity to bioaccumulate and have the potential for mutation and cancer, there has been a great deal of concern [167]. Ultimately, vaporization, photolysis, and enzymatic or microbiological destruction are possible outcomes for these heavy molecules [174].

Both on land and the ocean, crude oil spills have a major negative influence on the ecosystem. Oil disperses into tiny droplets in water, which can be creates persistent emulsions that harm ecosystems and marine life. The destiny of spilled oil on land is determined by the kind of soil and the activity of microorganisms; some components evaporate into the atmosphere, while others remain in the soil and pose a threat to living things and may result in long-term environmental harm. Because of their enduring nature and possible health risks, heavy chemicals like PAHs are especially worrisome. All things considered, oil spills require cautious control to reduce threats to the environment and public health. Related consequence of oil spilling is mentioned in Fig. 2.

4. Bioremediation of oil spills

Bioremediation has become the best and greenest method for eliminating or neutralizing pollutants [109]. The method by which native microorganisms naturally break down oil is known as

bioremediation. Numerous species, such as algae, fungi, and bacteria may break down oil in soils and/or water. Oil spill bioremediation could be the only cleanup method that can be carried out effectively and extensively [194]. Numerous substances, such as alkanes aromatics, and resins are found in crude oil. Many microorganisms such as *Pseudomonas*, *Rhodococcus*, *Acinetobacter*, *Gordonia*, *Burkholderia*, *Enterobacteria*, *Sphingobium*, *Novosphingobium*, *Sphingomonas*, *Psychrobacillus*, and *Bacillus*, were discovered in polluted locations. The ability of these bacteria to break down hydrocarbons from oil spills and other environmental pollutants is well recognized. They play an important role in bioremediation techniques [162]. Due to this restriction, efficient remediation necessitates a wide variety of species, and the relative concentration of each species varies during the bioremediation process. For example, organisms that degrade alkanes predominated right after the Deepwater Horizon leak. In the weeks after the spill, as the relative amounts of alkanes declined, the number of species that break down PAHs increased [76]. The ability of native species to digest oil and the speed at which communities can adjust to using oil as only a form of carbon or energy are the two main constraints for bioremediation. The amount of each species is determined by the kind and concentration of oil as well as environmental elements like temperature, salinity, climate, and availability of nutrients [151]. As a result, a variety of bioremediation strategies have been used in the area to quicken the pace of biodegradation and reduce the harmful effects of diesel fuel on local creatures. Many kinds of microorganisms that break down oil have been mentioned in Table 1. Bioremediation is further classified into two processes Bio-stimulation and Bio-augmentations. Remediation types are described in Fig. 3.

4.1. Bio-stimulation (BS)

Numerous elements, such as minerals, pH, humidity, temperature, oxygen, soil characteristics, and the presence of contaminants, might restrict the biodegradation of hydrocarbons in soil [17,22,49]. The process of changing the environment to encourage already-existing microorganisms that can perform bioremediation is known as BS. This can be achieved by adding different types of acceptors of electrons and restricting nutrients, which include carbon, nitrogen, phosphorus, and oxygen, which are ordinarily present in amounts that are insufficient to restrict the activity of microbial cells [172,178,81]. Characterization of the intrinsic degrading capabilities of the indigenous microbiota and the external factors involved in the kinetics of the in-situ process is necessary when using BS as a remediation strategy for petroleum pollutants removed from the soil. The principal benefit of BS lies in the fact that bioremediation will be carried out by naturally occurring microorganisms that previously existed, suitably adapted to the subsurface conditions, and geographically spread across the subsurface.

It has been demonstrated that the availability of the necessary nutrients in the polluted site might facilitate the breakdown of PH by a particular native bacterial community [71]. While nitrogen and phosphorus are scarce in most ecosystems, dispersed hydrocarbons from petroleum provide a significant supply of carbon for the local microbes. As a consequence, BS quickens the procedure of decontamination since it increases the bacteria population's capacity for degradation when several rate-limiting micronutrients are added to the system. Nikolopoulou, Kalogerakis [161,175]. The inclusion of BS is especially important in aquatic environments because, in addition to being directed towards the bacteria close to drops of oil in the saltwater, they also need to be resistant to being diluted and carried away by wave action. Furthermore, increased N and P substrate levels can lead to eutrophication, which promotes algae growth and eventually lowers the water's oxygen-dissolved content [161]. The level at which PHCs degrade in the natural environment is mostly determined by several variables other than nutrition. The degree of physical mixing, the use of mechanical tilling and shore washing as pre-treatments, and the accessibility of substitute carbon sources all have a significant impact on the

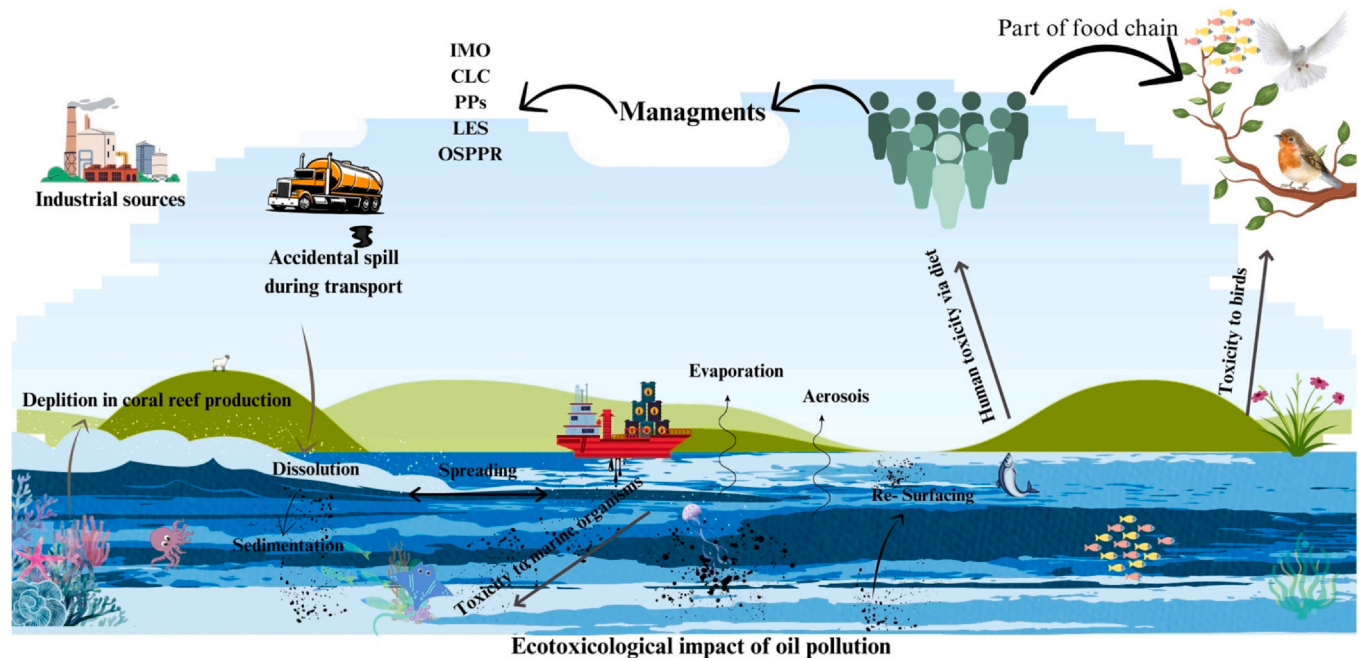


Fig. 2. Ecotoxicological impact of oil pollution.

capacity of microorganisms to mineralize oil following a spill. Oil may be dispersed into tiny particles by physical mixing, increasing its accessibility to microorganisms. By eliminating obstacles to oil availability, pre-treatments can either promote or inhibit microbial activity by upsetting microbial populations. Alternative carbon sources can either compete with oil for microbial attention or promote microbial development and oil decomposition [195]. The kinds of bacteria that flourish in a certain habitat depend on temperature. The ideal temperature ranges for various microbial species' development and metabolism differ. As a result, changes in temperature can affect the variety and makeup of the microbial populations that cause oil degradation [23]. In order to effectively breakdown pollutants such as total petroleum hydrocarbons (TPH), bio stimulation makes use of natural microorganisms that are already present in the subsurface environment and are suited to the local circumstances. There are several benefits to this approach:

The subsurface conditions of the area are well-suited to native microorganisms, which minimizes the need to introduce foreign species and may also minimize ecological impact. BS has the ability to remediate a large region of pollution without requiring much extra effort in microbial introduction because natural microorganisms are already dispersed across the subsurface. But there are also major obstacles to bio stimulation, especially when it comes to supplying nutrients and chemicals that encourage microbial activity:

Supplying supplements (such nutrients or electron acceptors) throughout the entire polluted subsurface is essential to the success of bio stimulation. Admixtures may be less available to the intended

microorganisms due to tight, impermeable lithology (such as clays), which might hinder their diffusion. Additives can follow preferred paths created by subsurface fractures, which might result in an uneven distribution and inefficient treatment of pollution throughout the affected region. Nutrients used to enhance biodegradation may unintentionally encourage the growth of heterotrophic microbes unsuited to break down TPH. Because the targeted microbial degraders may lose out on nutrients due to this competition for resources, BS may be less successful.

4.2. Bioaugmentation (BA)

Bioaugmentation involves adding specialized oil-degrading microorganisms to supplement native microbes in environments contaminated with oil. This approach is used when native microbes may lack the ability to fully degrade complex PH or are stressed after an oil spill. By introducing effective degraders, bioaugmentation aims to enhance degradation rates and efficiency, though success depends on factors like microbial compatibility, competition with native species, and sustained effectiveness over time [136]. Additional circumstances that might necessitate the use of BA include limited native hydrocarbon-degrading population, decontamination occurring quickly, and the potential for seeding to shorten the time it takes for the bioremediation process to begin [89]. The seed microorganisms in this approach need to be able to degrade the majority of petroleum components, preserve genetic stability and viability during storage, remain in unfavorable environments, effectively compete with native

Table 1
The common microorganisms that degrade oil offshore.

Group	Examples	References
Algae	<i>Agmenellum</i> , <i>Porphyridium</i> , <i>Amphora</i> , <i>Coccoliorise</i> , <i>Anabaena</i> , <i>Aphanocapsa</i> , <i>Oscillatoria</i> , <i>Nostoc</i> , <i>Chlorella</i> , <i>Chlamydomonas</i> , <i>Cylindrotheea</i> , <i>Dunaliella</i> , <i>Microcoleus</i> , <i>Petalonema</i> ,	[168]
Fungus	Yeast: <i>Acremonium</i> , <i>Pichia</i> <i>Candida</i> <i>Aspergillus</i> , <i>maltose</i> , <i>Aurebasidium</i> <i>Candida tropicalis</i> , <i>Cladosporium</i> . <i>Candida apicola</i> , <i>Candida</i> , <i>Debaryomyces</i> , <i>Trichoderma</i> , <i>Verticillium</i> , <i>Monilia</i> , <i>Mortierella</i> , <i>Rhodotorula</i> , <i>Saccharomyces</i> , <i>Torulopsis</i> Mold: <i>Geotrichum</i> , <i>Penicillium</i> , <i>Aspergillus</i> , <i>Gliocladium</i> , <i>Mucor</i> , <i>Fusarium</i>	[36,42]
Bacterium	<i>Achromobacter</i> , <i>Alcaligenes</i> , <i>Acinetobacter</i> , <i>Archrobacter</i> , <i>Actinomycetes</i> , <i>Bacillus</i> , <i>Cycloclasticus</i> , <i>Chromobacterium</i> , <i>Coryneforms</i> , <i>Nocardia</i> , <i>Flavobacterium</i> , <i>Micrococcus</i> , <i>Microbacterium</i> , <i>Vibrio</i> , <i>Mycobacterium</i> , <i>Pseudomonas</i> , <i>Sarcina</i> , <i>Serratia</i> , <i>Streptomyces</i> , <i>Xanthomonas</i> .	[36,95]

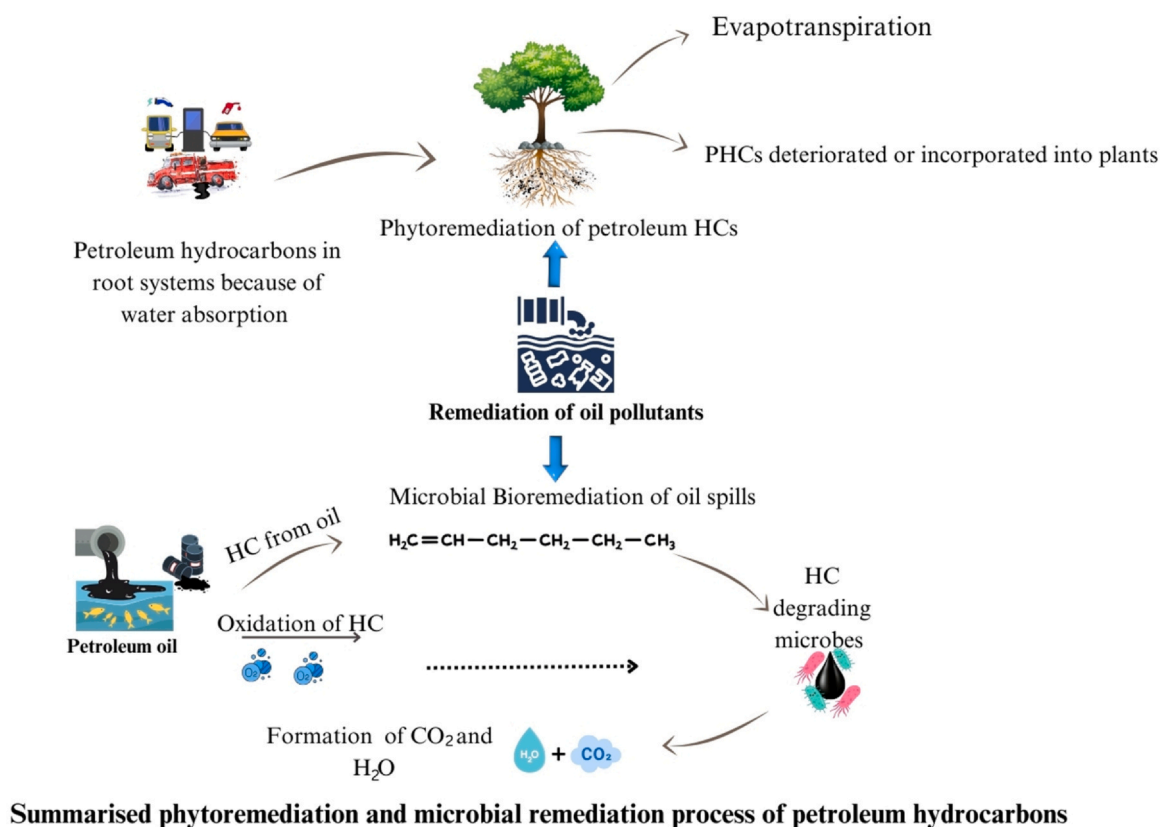


Fig. 3. Summarised phytoremediation and microbial remediation process of petroleum hydrocarbons.

microorganisms, and pass through the sediment's pores to the contaminants for the field method to be successful [98,36]. Varying species have varying preferences for breaking down different oil molecules, therefore microbial diversity and enzymatic capabilities are important factors in bioremediation. Certain microorganisms are specialized in mono- or polynuclear aromatics, whilst others focus on linear, branching, or cyclic alkanes. Selection of microorganisms with potential biotechnological applications in oil and petrochemical sectors is made easier by an understanding of their capabilities. The process of genetically engineering bacteria entails altering their genomes to improve their capacity to break down particular pollutants. By adding genes that encode enzymes specifically designed for bioremediation, this technique optimizes strains for specific uses. Through conjugation, biodegradation-related genes are packaged into vectors for transfer into native microorganisms, utilizing spontaneous genetic exchange processes to improve the capacity for local degradation. El Fantroussi, Agathos [78]. Prior understanding of the microbial communities residing at the target site should serve as the foundation for the most effective method of choosing competent microorganisms [199,203]. The capacity of the microbial community to break down the organic molecules may be hampered in co-contaminated locations, such as those with high metal concentrations and organic pollutants [179].

Both native and outsider bacteria that break down PH employ intracellular enzymes to convert the hydrocarbons into new sources of sustenance. Concentrated bacteria and oil-degrading microorganisms are grown on a growth medium comprising PH [209]. In bioreactors, microbes may occasionally overrun the spill site. This type of agent is meant to introduce a significant number of oil-degrading microorganisms into the impacted area, growing a community that breaks down oil to the extent that the oil spill is utilized as the primary energy source [186]. Scientists often utilize the genera *Pseudomonas aeruginosa* and *Bacillus subtilis* for BA. For the biodegradability test, scientists used oil-based hydrocarbons at various concentrations. Gasoline, Crude oil, kerosene, petroleum, diesel, lubricating oil, and other hydrocarbons

from petroleum were studied. In the process of the biodegradability test, the petroleum-based hydrocarbon rates varied from 0.5% to 5% [170]. The improved bacteria must have time to adjust to the pH, temperature, nutrition, and new petroleum-based oil that is accessible; nevertheless, other environmental variables may create unfavourable circumstances that prevent the oil from disintegrating [187]. The ambiguity surrounding the initial response clean-up method of the type bioremediation BA is partially caused by these elements and the undetermined timings of their acclimation phase. Water flow causes an ineffective water dilution, resulting in insufficient compounds, enzymes, and biosurfactants to break down the hydrocarbon molecular structure. A few chosen BA-BS studies for the degradation of PH are shown in Table 2.

4.3. Oil spill bioremediation mechanism

Both anaerobic and aerobic conditions can be used for petroleum oil bioremediation. Oxygen-oxidizing enzymes are required for aerobic metabolism because they convert O_2 into reduced substrate. Bacteria can address the difficulty of not coming into touch with water-insoluble hydrocarbons in two ways. A specific adhesion mechanism that produces emulsifying chemicals extracellularly improves the contact [118]. Several PH models are often used to study the mechanism of oil spill deterioration. However, in most cases, the attack of microorganisms on the petroleum structure initiates a series of sequential processes that lead to the biodegradation of PH and the creation of intermediate compounds.

A variety of microbes use intermediate chemicals, which cause additional breakdown.

Jafarinejad [117]. The hydroxyl group is appended to the end of the alkane chain in the [150] initial stage of the degradation pathway. This group can combine with the unsaturated ring of PAH to produce alcohol. The component then oxidizes to aldehyde and then carboxylic acid, reducing the length of the chain. Ultimately, biomass, CO_2 , and

Table 2

A compilation of documents that focus on the use of bioaugmentation and bio-stimulation to degrade petroleum hydrocarbons.

Micro-Organisms	Pollutant	References
<i>Alcanivorax</i> , <i>Thalassolituus</i> , <i>Cycloclasticus</i> <i>Pseudomonas aeruginosa</i> AspH2	0.1% (v/v) weathered crude oil in seawater in 30 days	[4]
<i>Alcanivorax borkumensis</i> SK2 <i>Pseudomonas</i> <i>Pseudomonas aeruginosa</i> , <i>Aspergillus niger</i> <i>Alcanivorax borkumensis</i> , <i>Marinobacter hydrocarbonoclasticus</i> , <i>Alcanivorax dieselolei</i> , <i>Thalassolituus oleivorans</i> , <i>Cycloclasticus</i> sp. 78-ME <i>Proteobacteria</i>	nC15–nC35 (TPH = 10 g/L) in 30 days 1000 ppm polluted seawater in 20 days 0.5% (w/v) crude oil in 28 days 10% (v/v) Crude oil in 8 weeks 2% (v/v) Cargo fuel oily seawater in 14 days	[82] [104] [79] [18] [64]
	1% (v/v) Diesel oil in 7 days	[64]

Table 3

Few examples of phytoremediation plants to eliminate pollutants using different substrates.

Plant	Substrates	References
<i>Callitriche lusitanica</i>	Aquatic	[84]
<i>Eleocharis acicularis</i>	Aquatic	[183]
<i>Fontinalis antipyretica</i>	Aquatic	[84]
<i>Spirodela polyrrhiza</i>	Aquatic	[74]
<i>Helianthus annuus</i>	Soil	[7]
<i>Zea mays</i>	Soil	[111]
<i>Canna generalis</i>	Soil	[43]
<i>Medicago sativa</i>	Soil	[220]

H₂O are produced. Hydrocarbons become more polar, water-soluble, biodegradable, and less poisonous when oxygen is added to their structure [117]. First, alcohols are created when aliphatic hydrocarbons like n-alkanes degrade. These alcohols are Hthen oxidized and dehydrogenated one after the other to yield primary alcohols, aldehydes, and monocarboxylic acids. Following β-oxidation, the carboxylic acids generate fatty acids and acetyl coenzyme while releasing carbon dioxide [118]. The addition of oxygen to the hydrocarbon is the limiting step, and once carboxylic acid is created, it may be eliminated quickly. With branched isoprenoid alkanes, like pristine the hydrocarbon oxidizes to produce dicarboxylic acids. The susceptibility of hydrocarbons to microbial attack is increased when methyl branches are present [34]. The process by which aromatic and PAHs are broken down involves the hydroxylation of the ring by mono- or dioxygenase enzymes. As a result, diol is produced, the ring is broken, and more deterioration occurs [118].

5. Phytoremediation

Another bioremediation process method that could potentially be utilized as a substitute for the metal-based removal process is phytoremediation. Metal-accumulating vegetation is used in the environmentally and economically advantageous "green" process known as phytoremediation of metals to eliminate organic pollutants and dangerous metals, especially radionuclides, from polluted soils and water [13]. A sustainable and environmentally friendly method of eliminating or breaking down pollutants from the environment is referred to as phytoremediation. Rhizoremediation, the intricate interactions of root systems, roots exudate, rhizosphere, and microorganisms, enables plants to mediate multiple processes in the removal of oil spills from damaged settings through a remediation procedure.

In addition to the soil environment, higher vegetation root structures interact with an additional population of metabolically active microbes. Along with the roots, the live plants provide a unique ecosystem where the number of microorganisms is far higher than in a soil environment without roots [143]. The soil produces approximately forty percent of the carbohydrates, organic acids, and aromatic

compounds that plants exude during photosynthetic rooting. These compounds are abundant in carbon and provide energy for microbial development. These exudates trigger the chemotactic signal in bacteria that enables them to go nearer the roots and create root colonization, which in turn stimulates the development and activity of bacteria responsible for the breakdown of organic contaminants [27]. The unique chemical structures and exudation rates of various species have diverse effects on microbes; hence, the specific component of plant exudates contributes to the degrading activity [31]. In addition to producing oxygen for rhizosphere bacteria, plant roots can help enhance the oxidative breakdown of hydrocarbons by allowing oxygen to permeate the soil and improving soil structure. carbon dioxide, Alcohol, acids, and water are the breakdown's final products; these are less hazardous and less durable than the original components [94]. Table 3 Few examples of phytoremediation plants to eliminate pollutants using different substrates.

5.1. Phytoremediation of oil spillage in the soil system

At first, plants showed good resistance to soil tainted with crude oil. For example, the 4 o'clock flower (*Mirabilis jalapa* L.), whose capacity to withstand contamination from crude oil, has been successfully proven as a phytoremediator [192]. Over 127 days, *M. jalapa* increased the removal effectiveness of TPHs [193]. Certain vegetation's physiological and morphological traits enable its roots to draw more microbes and promote the breakdown of hydrocarbons [21]. It has been found that the phenolic chemicals and flavonoids released from the roots of mulberry (*Morus spp.*), Osage orange (*Maclura pomifera*), and apple (*Malus domestica*), trees activate PAH-degrading microorganisms. It has been suggested that garden balsam (*Impatiens balsamina* L.), an attractive shrub, may be useful in effectively removing oil from polluted soils. The primary factor in oil deterioration was the large rise in the overall number of live microorganisms surrounding the plant root over the four-month cultivation period [50]. Certain Forest tree types, such as gmelina (*Gmelina arborea*) and teak (*Tectona grandis*), have demonstrated adequate capacities to flourish well in a polluted environment with crude oil up to ten percent w/w of soil. At greater oil exposure levels, however, the study plants' growth and productivity were considerably impacted [6]. It has also been demonstrated that the native Brazilian tree known as branquilha (*Sebastiania commersoniana*) is resistant to petroleum pollution in the soil. In polluted soil, this tree reduced petroleum-based hydrocarbons, by up to 94% [200]. Seven plant types were assessed for seed germination and early development in testing environments polluted with crude oil These kinds of plants included lettuce (*Lactuca sativa*), corn (*Zea mays*), millet (*Panicum miliaceum*), watermelon (*Citrullus lanatus*), sorghum (*Sorghum bicolor*), soybean (*Glycine max*), and okra (*Abelmoschus esculents*). The findings showed that in contrast to other plants millet, sorghum, okra, and corn had a high resistance to oil phytotoxicity [166]. *Zea mays* and soybeans *G. max*, 2 kinds of crops, have also shown tolerance to soil polluted with crude oil [115].

Another feature of plants that makes them appropriate for phytoremediation is their capacity for petroleum-based hydrocarbon biological accumulation. The canna lily (*Canna indica* L.) shoots have a significant accumulation of benzene, toluene, ethylbenzene, and xylenes BTEX [43]. In just 21 days, canna eliminated 80% of the BTEX in the growing zone soil. In soils polluted with toxic metals and petroleum products, the tropical attractive shrub known as Siam weed (*Chromolaena odorata* L.) demonstrated a significant capacity for phytoaccumulation. From soil contaminated with toxic substances and petroleum, these species were able to extract up to eighty percent of the crude oil. Tall fescue and perennial ryegrass, two perennial types of grass with broad root networks and strong growth following planting, have also been chosen for oil phytoremediation [62]. Plant species having tap root structures, as opposed to surface root structures, might influence toxins that are situated deeper by reaching beneath the soil or the water column [123]. It has proved effective in employing deep-rooted plants, such as *Salix* spp. and *Populus* spp., to absorb water from underground that contains TPH [85].

5.2. Phytoremediation of crude oil spills in aquatic ecosystem

Different plant species known as macrophytes—which can be emergent, submergent, or floating—can be found growing in or next to water in aquatic environments including rivers, lakes, and wetlands [40]. They could be utilized as phytoremediators of oil hydrocarbons. Their rapid growth is one of the traits that makes them appropriate for phytoremediation. They spread quickly and are intrusive. As a result, they may be rapidly replaced by new growth following the harm caused by oil contamination [40]. Certain aquatic vegetation has fibrous roots that provide greater surface area and thicker rhizospheres for the establishment of microorganisms [218]. The dense root systems of *Eichhornia crassipes* can greatly reduce the amount of flowing petroleum-based hydrocarbons in surface waterways [159]. Additionally, biscuit grasses were mentioned as possible alternatives for phytoremediation using PH. Their root structure allows them to grow and survive in soils polluted with diesel (up to 30 g.kg⁻¹) [185]. In addition to their robust vitality and large root surface area, reeds are dominating coastal wetland vegetation that can aid in the restoration of wetlands damaged by petroleum (J. [214]). Alligator weed (*Alternanthera philoxeroides*), common reed (*Phragmites australis*), maidencane (*Panicum hemitomon*), and duck potato (*Sagittaria lancifolia*) are four kinds of fresh-marsh vegetation that efficiently perform phytoremediation [75]. Based on the anoxic and hypoxic states of the sediments and soils in aquatic environments, Crude oil undergoes anaerobic breakdown, a process that is extremely slow and unfinished. Certain macrophytes enhance the aerobic respiration process of rhizosphere microorganisms by moving ambient oxygen from the shoots to the roots ([171]b; [154]). Wetland plants, or submerged freshwater macrophytes, have a natural defence mechanism that allows them to oxygenate their root region and defend against phytotoxins like Fe²⁺, H₂S, and Mg²⁺ [171]. Phytoremediators can be used for floating species because of their capacity to bioaccumulate and biosorb contaminants from the fluid medium, even in cases when the plant's root system does not form into a rigid matrix [153,176]. Investigations on the removal of crude oils from oil-polluted water bodies have been conducted using floating plants, which include water lettuce (*Pistia stratiotes* Linn.) and duckweed (*Spirodela polyrrhiza* Trev.). But the outcome was unimpressive [5,10]. Studies determining aquatic species' capacity for petroleum-based phytoremediation are often few. It would appear vital to examine the aquatic macrophytes' effectiveness because the majority of oil spills happen in aquatic areas.F.

6. Remediation through natural sorbents

Natural sorbents are widely available and provide several significant benefits. Moreover, the substance itself decomposes naturally, providing further advantages for the environment [91]. Furthermore,

sorbents derived from natural materials have significant hydrophobic, oleophilic qualities. Natural sorbents include things like peels of bananas, kapok pulp, hair from humans, and fibers such as wool and cotton [113,145,169,212,220,61,80]. Tropical fibers are employed as sorbents in the extraction of toxic hydrocarbons, including petroleum and its distillates (such as kerosene, gasoline, and creosote,) and the extremely dangerous pentachlorophenol. In this instance, fiber materials made from the leftovers of tropical plants (palm, banana, coconut, and pineapple) were treated with an alum solution to increase their hydrophobicity. A hot compression procedure might be used to extract the oil from the sorbent substance [106]. A different type of trademarked sorbent device makes use of chitin polymer, a type of glucose compound. The apparatus is put together and has a cotton fabric cover. It is held aloft by a plastic or hardwood flotation mechanism. Oil can seep through the porous cotton fibers and get absorbed by the chitin granules in the sorbent base [66]. The particular ball-shaped kenaf sorbent, measuring between 0.2 to 10 g, was shown to have a capacity for absorption of up to 1000% and possibly 1800% of its initial weight in certain instances. These kenaf grass balls might be burnt or thrown in the trash after use. To create an upsurge in oil recovery, naturally occurring substances including, paper pulp, bagasse, sawdust, chicken feathers, peat, and straw are used. Unaltered naturally organic absorbents have a relatively low reusability rate, which suggests that their cost and price may rise [107]. Furthermore, natural organically absorbent materials have a very low resilience and may even break or deform when used. For instance, appliances based on native organic sorbent were recommended for use in tiny oil spills when the water surface is in control. Particularly, a human hair by-product shows great promise as a low-value waste byproduct that can be used as a natural sorbent for crude oil and other contaminants mentioned in Table 4. This represents a creative new use for an uncommon resource-based product that the human population continuously produces [112,20]. Using human hair, a boom will be constructed to contain and lessen the effects of crude oil spills. To determine how successfully human hair waste-based booms absorbed spilled oil and saltwater while maintaining their buoyancy, researchers compared them to other natural sorbents such as textile byproducts, reprocessed cellulose, and industrial polypropylene-based booms [164]. In short, natural biological sorbents are an intriguing material to combine with the growth in oil spill cleaning because of their affordable, readily available, eco-friendly, and biodegradable qualities. A diagrammatic representation mentioned in Fig. 4.

7. Physical method of oil spill remediation

One of the most effective and ecologically responsible techniques for cleaning up oil spills. Since physical procedures don't include the use of any chemicals, government authorities don't need to be consulted in advance in order to perform cleanup operations. Sorbents, oil booms, and skimmer are the techniques most often employed for remediating maritime oil spills. All physical remediation techniques has been present in Fig. 5.

7.1. Booms

Oil will float on the water's top since it is often less thick than water. Booms are fixed floating apparatus used to stop this oil from spreading. Several countries all around the world have used the use of booms as a main maritime response tactic to contain the spreading of oil [72]. Physical obstacles known as booms stop the oil slick from spreading. To do additional cleanup procedures, they contain the oil and keep it at a sufficient thickness. Mapelli et al., [146]. Booms are primarily used to contain oil and prevent it from spreading. However, they can also be employed to prevent an oil slick from reaching shorelines, areas that are expensive and difficult to clean up, or biodiversity hotspots, so safeguarding such areas. Three physical characteristics—buoyancy, roll response, and heave response—are essential to a boom's functionality

Table 4
The table provides an overview of the many kinds of natural sorbents that are utilized in oil remediation.

Type of Natural Sorbent	Source Material	Function	Advantage	Disadvantage	References
Peat Moss	Peat bogs	Absorbs and retains oil	High absorbency, biodegradable, widely available	Can become waterlogged, limited reuse	[181]
Straw	Agricultural byproduct	Absorbs oil, forms mat on the water surface	Inexpensive, readily available, biodegradable	Requires large quantities, limited reuse	[126]
Cotton	Cotton plants	Absorbs oil, lightweight	High absorbency, biodegradable, renewable	Can become waterlogged, less effective for heavy oils	[11]
Coconut Coir	Coconut husks	Absorbs oil, forms mat	Renewable, biodegradable, high absorbency	Limited reuse, requires processing	[181]
Kenaf	Kenaf plants	Absorbs oil, lightweight	High absorbency, biodegradable, renewable	Limited reuse, requires processing	[208]
Activated Carbon	Biomass (e.g., coconut shells)	Absorbs oil and hydrocarbons	High adsorption capacity, reusable, effective for various oils	Higher cost requires regeneration	[4]

[19]. The efficiency of the boom is significantly influenced by the force of buoyancy on it. By keeping the boom afloat on the water's surface, buoyancy assists to maintain the boom's freedom of motion above the water's surface [19]. As boom weight falls and buoyant force increases on the boom, heave response rises. Booms that have a good heave response or a sufficient length of freeboard are better able to stop the spread of oil slicks caused by wave motion. In comparison to booms with sufficient freeboard length, those with superior heave response are more effective at containing oils. Effectively tracking the sea surface is enhanced by a boom with strong flexibility or heave response. Booms may be made more flexible by reducing the length of their float portions and the distance between them. Additionally, foam pellets in the inflated flotation chamber increase the booms' flexibility so they can more readily follow wave motion and stop oil slicks from spreading because of the water's vertical motion [19].

7.2. Skimmers

Skimmers are among those gadgets that are nearly usually utilized in tandem with booms. Depending on the oil spill, they can be autonomously mobile or fixed, and run from ships or land. When collecting and pumping oil, skimmers are physical/mechanical devices that do not alter the oil's qualities. In contrast, chemical and thermal processes alter the oil's fundamental structure, making it worthless. Reprocessing recovered oil is feasible but requires careful selection of the appropriate skimmer. Key factors include the type of water body (sheltered vs. open sea), oil characteristics (viscosity, adhesive properties, weathering), and environmental conditions (wind speed, wave strength, debris level, presence of ice or seaweeds). These considerations ensure efficient oil recovery and minimal environmental impact. Skimmers are most effective with a significant thickness of oil slick, but their efficiency decreases as the slick thins. Weathering alters the oil's characteristics over time, requiring different skimmers to manage various types of oil spills effectively.

7.3. Sorbent materials

The most common naturally occurring item used to clean up oil spills is sorbent material. Vermiculite, zeolites, activated carbon, organo-clays, glass, sand, and volcanic ash are examples of common natural inorganic sorbents [51]. Natural inorganic sorbents like Ekoperl and Pevatone are significantly superior to previously used materials such as vermiculite, zeolites, and activated carbon. Despite their effectiveness, the high cost of Ekoperl and Pevatone has limited their widespread use [185]. Mineral inorganic sorbent materials are stable both chemically and physically, and they provide a large surface area for sorption. Since they are made of natural raw materials, they are affordable and effective solutions for cleaning up oil spills on the sea. Mineral sorbents, in contrast to natural/organic sorbents, are typically non-flammable and resistant to both basic and acidic chemicals [29,33]. Mineral inorganic sorbent materials can absorb oil weights four to twenty times their weight, demonstrating a respectable absorption capability [196]. The polymeric materials that are most frequently utilized in the production of synthetic sorbents include polyester, polyvinyl chloride (PVC), polyethylene (PE), polyurethane (PU), polypropylene, and polystyrene (PS) [16,125]. Every synthetic adsorbent has a unique pore structure and porosity. Their high water-to-oil intake is a result of their greatly improved hydrophobic and oleophilic properties as compared to their counterparts in the organic and mineral domains. Most synthetic adsorbents can absorb oil weights 70–100 times their weight because of this enhanced capacity; open-cell polyurethane sorbents may absorb oil up to 100 times their initial weight [72]. Sorbents composed of polypropylene are recognized for their superior ability in oil adsorption [198].

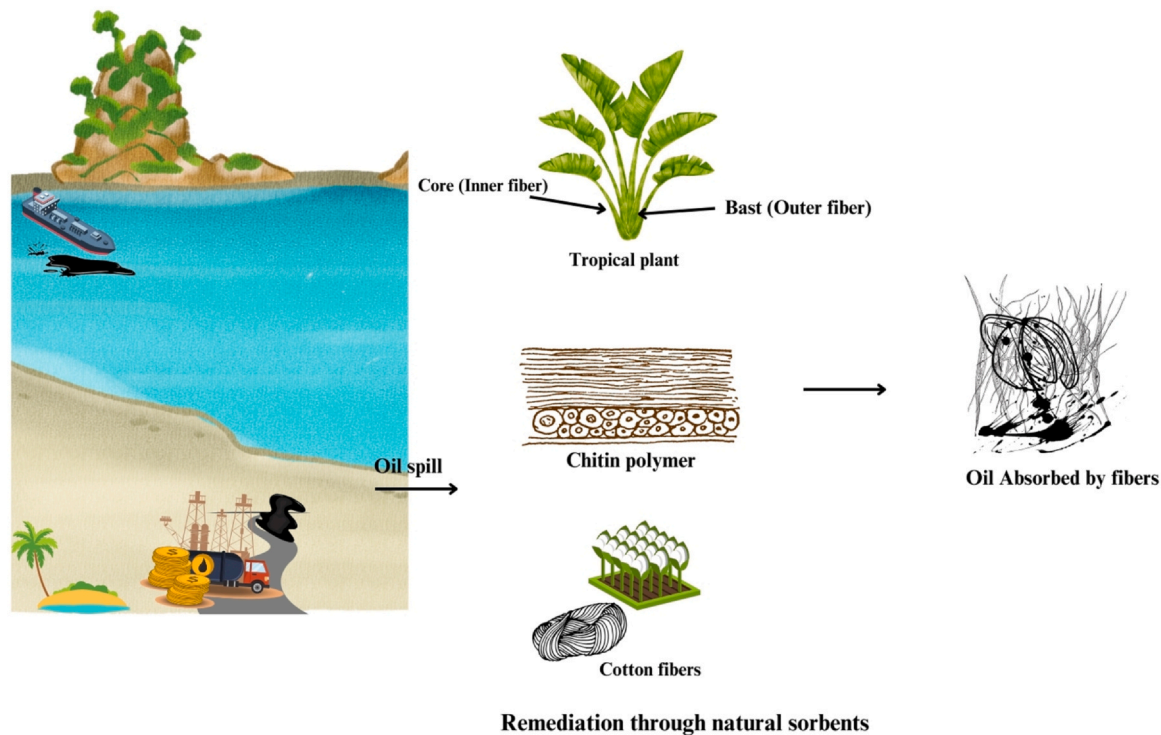


Fig. 4. Remediation through natural sorbents.

8. Chemical remediation by natural environmental process

Environmental chemical processes like oxidation and reduction have the power to change Mineral inorganic sorbent materials can absorb oil weights four to twenty times their weight, demonstrating a respectable absorption capability. pollutants into less harmful forms. For instance, organic molecules can undergo oxidation by airborne oxygen, which lessens their toxicity. Oil can undergo photooxidation or autoxidation [26]. Hydrocarbons are changed into hydroperoxides by free-radical transformations that are sparked by sunlight. After that, these hydroperoxides undergo further transformations to produce acids, ethanol, and other oxygenated substances. The partly oxidized

hydrocarbons polymerize as a result of the free-radical processes. Compared to the source hydrocarbons, the resultant "tar" is thicker, more polar, and more viscous [173]. Temperature, sunshine intensity, and the physical condition of the oil all affect how quickly autoxidation occurs. Oil-thin films and emulsions in water have a higher probability of autoxidation than big coherent bulk [229]. Seawater minerals can act as autoxidation catalysts. Autoxidation is also catalysed by some metallic elements found in crude oil, which include vanadium porphyrins [229]. Conversely, petroleum's sulphur-containing molecules function as antioxidants and prevent autoxidation. Thus, compared to low-sulphur oil-based products, high-sulphur oils like Kuwaiti oil are more resistant to autoxidation. Crude oils will most easily attack

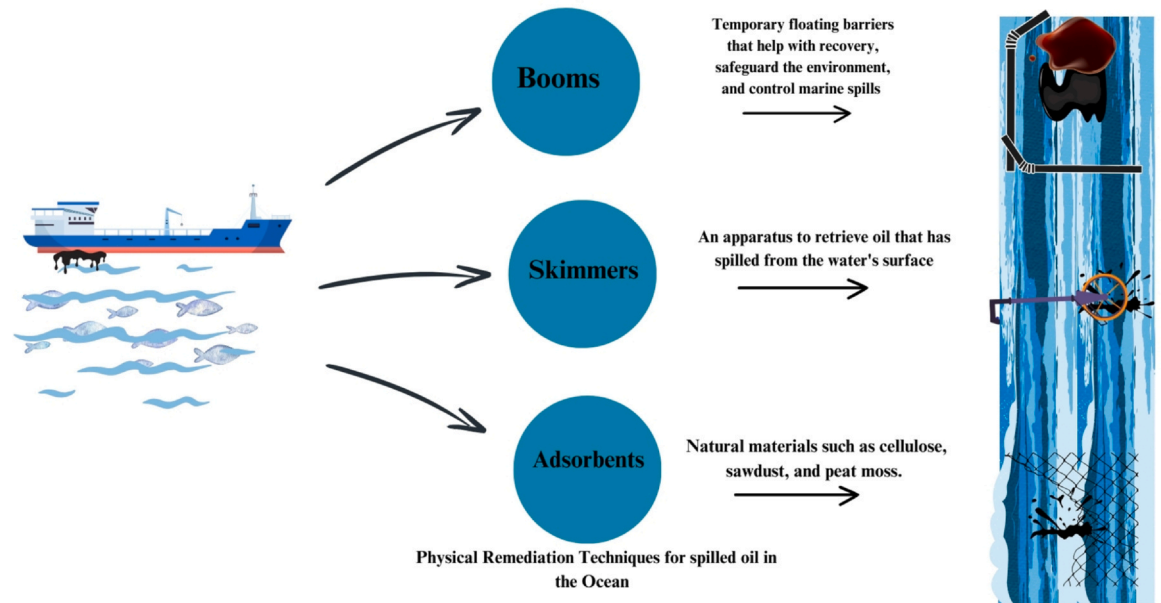


Fig. 5. Physical Remediation Techniques for spilled oil in the Ocean.

Table 5
Table summarizing the different types of chemical remediation methods for oil spills.

Method	Mechanism	Advantages	Limitations	References
Dispersants	Chemically dissolving oil into tiny droplets	treating sizable afflicted regions, clearing surface oil, promoting biodegradation,	Low efficiency and highly weathered emulsified oils harm marine animals like plankton and are ineffective in calming saltwater.	[4]
Solidifiers	Oil-solidifying gelling agents	Stop oil from spreading by immobilizing it, solidifying it quickly, and making it easy to apply.	It is not appropriate to use many solidifiers on wastes made from viscous oil that contains volatile hydrocarbons.	[189,156]
Emulsifiers	Demulsification	Reduced concentrations, quick separation of oil and water, increased oil collecting efficiency.	Toxic to marine life	[228,119]
Clay minerals	Clays encapsulate oil droplets, which hinders their diffusion and increases their accessibility to microbial decomposition.	Large Surface Area, Cation Exchange Capacity, Cost-Effective	Adding clay minerals to aquatic environments may cause sediment disturbance, which may release additional pollutants.	[70,11]
Biochar	Oil spill chemical cleanup is gaining traction as a possible use for the pyrolysis of organic materials, which yields a carbon-rich product.	High Adsorption Capacity, Environmental Benefits, Improvement of soil quality	Removing biochar from polluted areas can be difficult once it has absorbed oil, particularly in aquatic settings.	[158,11]

paraffin and aromatics with the right-side chains. Straight-chain alkanes will not autoxidize as quickly as iso-alkanes. Rapid autoxidation can occur in hydrocarbons with resonance-stabilized alkyl radicals, such as cumene and tetralin. For oil removal at typical seawater temperatures, autoxidation does not have a significant impact mentioned in Table 5.

When it comes to chemical remediation, advanced oxidation processes (AOPs) are thought to be more efficient and cost-effective than physical and biological procedures since they change organic contaminants into less hazardous or harmless compounds [227]. AOPs include ozone, the oxidizing agent hydrogen peroxide, persulfate, non-thermal plasma, and other oxidation systems. The oxidants that make AOPs can include ozone (O₃), persulfate (S₂O₈), and hydrogen peroxide (H₂O₂) [57,129,139,222,223]. Ozone is simpler to travel in contaminated soil than fluid reagents, which is why it is frequently employed for in situ remediation of soil with low moisture, coarse soil particles, and low soil organic matter (SOM).

There are two ways that ozone oxidation eliminates TPHs from soil [184]. One is contaminants having either electrophilic or nucleophilic qualities, ozone directly oxidizes electron-dense chemical connections or nuclei of atoms [223]. Second is under the influence of SOM or metal oxide, ozone breaks down to produce radicals, which oxidize organic contaminants into tiny molecular molecules [213]. With its powerful oxidation capacity, the ozone oxidation system has already seen extensive use. Ozone can greatly increase the hydrophilic properties and degradation of PH, particularly those with aromatic rings and complex constituents [225,224]. The features of TPHs determine the ideal circumstances for the subsequent stages of ozonation and biological degradation. Before and post-ozonation were similarly efficient at degrading TPHs for the components that were appropriately biodegradation.

9. Factor affecting remediation of oil spillage

Numerous variables affect the biodegradation of organic molecules, particularly PH. The rate and extent of biodegradation are influenced by several biological and physical factors, including the presence of particular microbial communities that are capable of metabolizing PH, temperature, pH, and moisture levels, as well as the size, structure, and chemical makeup of the PH molecules. The biodegradation process can also be impacted by the availability of oxygen and nutrients, as well as by the presence of any compounds that impede it. In general, knowing these factors is essential to developing efficient bioremediation plans for polluted areas [38,53]. Therefore, when choosing a suitable remediation technique, the PH composition of the affected locations should be the primary consideration. Leaked crude petroleum oil often creates an emulsion that enhances its surface area and makes it more bioavailable, but big emulsion masses and tar-balls—aggregates of weathered and undegraded oil—hinder biodegradation because of their smaller surface area [134]. Environmental elements including salinity, temperature, pH, oxygen content, and nutritional value also affect PH biodegradation [207,24]. Fig. 6 provides a summary. Environmental elements influencing how oil's PHCs break down.

9.1. Temperature

One physical component that primarily influences the biodegradation of PH is **temperature**. At low temperatures, temperature changes the physiology and variety of microorganisms, it can make cleaning more challenging by slowing down evaporation, solidifying some forms of oil, and reducing microbial activity. At high temperatures, temperature increases fluctuation and viscosity [210]. Elevated temperatures can potentially accelerate oil's evaporation and microbiological deterioration.



Fig. 6. Environmental factors affecting the breakdown of PHCs of oil.

9.2. Oxygen

When oxygen levels are low, nutrient supplementation is not anticipated to significantly increase the rates at which oil biodegrades [92]. As the primary electron acceptor in aerobic bioremediation, oxygen deficiency can restrict the ability of aerobic bacteria to biodegrade. Anaerobic bioremediation, which is slower than aerobic bioremediation, takes over as the predominant activity when oxygen is absent [180]. Maintaining and advancing aerobic bioremediation might have a relatively high oxygen requirement. Regarding hydrocarbon pollutants, many studies calculate varying stoichiometric ratios of oxygen needed for the pollutants to completely degrade: For every mole of hydrocarbon, 3 M O₂, 3 kg of oxygen, 8.6 moles of oxygen for every mole of degraded hydrocarbon, and 0.5–1 g of oxygen/g of hydrocarbon are all given [88,90,97]. However, these criteria are highly challenging to achieve and can only be loosely anticipated due to the changing and dynamic characteristics of the maritime environment. Supplying oxygen is necessary for bioremediation procedures and can accelerate the breakdown of PH, even the heavier, more persistent molecules. For the native microbiome to continue its aerobic biodegradation, there must be a sufficient flow of oxygen.

9.3. Inorganic nutrient concentrations

Inorganic nutrient concentrations are often low in aquatic settings. On the other hand, PH, rich in carbon, nitrogen, and phosphorus in varying ratios, is released during oil spills and leaks, creating an environment that is not conducive to microbial development. Atlas [25,63]. On the other, hand, biodegradation in freshwater wetlands with PH contamination requires nutritional supplements [60]. The makeup of bacteria and fungi greatly influences the breakdown of PH-contaminated matrices, as these microorganisms can only metabolize a restricted range of hydrocarbon substrates. Britton [48]. PH-degrading bacteria create gases, surfactants, biological polymers, and organic chemicals that aid in improved bioremediation and crude oil recovery [135,182,37]. **Microbe-secreted biosurfactants** function as emulsifying agents by lowering oil's surface tension and creating micelles. The oil droplets encased by biosurfactants are extremely accessible to bacteria, allowing for further breakdown [54].

Primarily, research on the variety of microbes in PH-contaminated environments and the impact of environmental variables on microbes' capacity to biodegrade would help to enhance management techniques for the effectiveness and pace of bioremediation in the contaminated areas [1]. Due to constraints like identifying and isolating only culturable microorganisms, metagenomic techniques are being used [99]. The metagenomic study offers a thorough picture of the community of bacteria and the functional genes responsible for PH pollution adaption and PH breakdown of both aromatic and aliphatic compounds [1]. It is also important to note that, since PH-contaminated spots are also inadequate in vital nutrients like nitrogen, phosphorus, sulphur, and iron, attention must be paid to genes involved in **heavy metal resistance** and nutrient metabolism in addition to the documentation of a variety of catabolic gene sequences for aliphatic and aromatic PH degradation [120]. These genes' analysis will be a useful tool for creating ideal ecological and physicochemical conditions to increase the rate and effectiveness of biodegradation. Most microorganisms use PH as a source of carbon and energy through a cellular metabolic process. Additionally, in the presence of PH, bacteria exhibit a variety of, physiological, cellular, and biochemical changes. Less research has been done on how this adaptation affects microbial activity, physiological reactions to PH, and susceptibility to and biodegradation of PH [205].

9.4. Microorganism

Hydrocarbon-degrading microorganisms are often found in the maritime environment, where they consume spilled hydrocarbons as an energy source to fuel their rapid growth and replication. Furthermore, alterations in some catalysts and other physiological elements in contaminated marine environments raise the number of microorganisms that break down hydrocarbons [133]. Therefore, in the PH-polluted marine ecosystem, the overgrowth of hydrocarbon-degrading microorganisms affects the regular organization of the microbial community [211]. PH, a kind of lipophilic chemical, is highly attracted to hydrophobic environments, such as the lipid bilayer of cell membranes. PH molecules can damage the integrity and fluidity of a membrane when they build up inside its hydrophobic area. This disruption can have several impacts on how cells operate, such as changes in the fluidity of the membrane, modifications to the structure of membrane proteins,

and suppression of membrane-associated enzyme activity. The regular operation of membrane-bound transport proteins, enzymes, and receptors can be hampered by the presence of PH in the membrane, which eventually compromises the survival and functionality of cells. Furthermore, PH-induced membrane damage might impair a cell's capacity to respond to outside stimuli and maintain homeostasis [142]. PH do not specifically interact with individual biological components; instead, they predominantly interact with bacterial cells through passive transport pathways. The term "passive transport" describes the flow of molecules across a cell membrane that is not dependent on the cell for energy input, but rather on concentration gradients [44]. Some bacterial cells create biofilms, which change the hydrophobic state of their cell surface to function as a physical barrier for PH diffusion into the membranes, preventing these passive interactions. Certain bacteria have evolved systems that use specialized proteins called efflux pumps to actively remove PH from their cytoplasmic membranes. These membrane transporters, known as efflux pumps, actively move hydrophobic substances such as PH out of the cell to lower their intracellular concentration [114]. An intriguing phenomenon seen in many bacteria is chemotaxis. Bacteria may travel toward or away from particular compounds in their environment, such as PH, thanks to a behavioral response. On the surface of the bacterial cell, specific proteins known as chemoreceptors work to facilitate this movement. Some chemoreceptors on the surface of bacteria can bind to PH or other chemical substances found in the surrounding environment. The bacterial migration toward or away from the chemical gradient is caused by alterations in the direction of flagellar rotation, which are eventually brought about by this binding and a signalling cascade inside the cell. More research on microorganisms' taxis related to PH will reveal more about the defensive behaviour of various genera toward varying PH, the recognition of putative chemical receptors through genome sequencing and communication, and the wide range of PH-metabolizing organisms through signal translation in chemical receptors generated by hydrocarbons. Furthermore, microorganisms exposed to petroleum modify their genetic pathways in a way that amplifies the genes or genes involved in PH metabolism. The development of new microorganisms with enhanced PH metabolism will be aided by the development of methods for detecting catabolic genes [103]. Although the microorganisms found in PH-affected sites soils in particular will soon undergo these alterations in behaviour and adaptation.

9.5. pH

The majority of heterotrophic bacteria and fungi favor neutral pH settings, although terrestrial habitats are more erratic than aquatic ecosystems because of the chemistry of the soil, which may cause pH variations of 2.5 to 11.0 [9]. The ideal pH range for hydrocarbon biodegradation is 6.5–8.0. Fertilizer applications and the aliphatic acid generation that results from alkane biodegradation can both reduce the pH of soil [9]. For an alkaline or acidic correction, respectively, added lime or sulfur will modify the pH. The susceptibility of aromatic degraders to pH is the cause of aromatic hydrocarbons' longevity in the environment [149]. When pH was adjusted from 7.0 or 9.0 to the ideal pH of 8.0, the biodegradation rate constants for several PAHs doubled (B. V [55]).

9.6. Salinity

The prevailing consensus is that there is an inverse relationship between salinity and the biodegradation of PH, despite the existence of bacteria known as halophiles that require salt for development. Margesin, Schinner [149]. On the other hand, nothing is known about how salinity affects the bioremediation of soil polluted by petroleum. According to one research, adding artificial salinity up to the same level as an oil-field brine reduced the rate of oil deterioration by 20–44% [149]. More than 1% (w/v) of salt slowed down the rate of

mineralization and lengthened the lag period before weathered hydrocarbons from production facilities biodegraded. In certain situations, the amount of mineralization was even reduced. concentrations of salt between 0.5% and 1% (w/v) increased microbial activity. Ulrich et al., [202]. Acclimation to high salt concentrations may lessen the effects on microorganisms present in naturally saline soil conditions [149].

10. Case studies

The 1989 Exxon Valdez oil disaster was a pivotal moment in environmental history. About 11 million gallons of crude oil leaked into the pristine waters of Prince William Sound, Alaska, when the oil ship Exxon Valdez came aground on Bligh Reef. The accident spilled approximately 41000 m³ of crude oil and contaminated about 2000 km of coastline. An essential component of the cleaning operations after the Exxon Valdez accident was bioremediation. Utilizing microorganisms, this procedure neutralizes or breaks down environmental contaminants. Certain microorganisms were used in the oil spill situation to naturally break down the oil, reducing the negative effects on the ecosystem. Bioremediation was applied widely. The injection of nutrients was utilized in coastal areas, such as wetlands and beaches [219,46]. Afterward the Valdez spill, research on the use of fertilizers on a particular beach revealed a five-fold increase in oil deterioration [46]. Furthermore, certain research and initiatives in Alaska have revealed that commercial fertilizers may be just as successful as commercial bioproducts in accelerating biodegradation. Furthermore, for bioaugmentation projects, fertilizers could be a more affordable option than specialist bioproducts [45].

Fertilizer has a considerable effect on the three-year in-situ oil degradation in the soil of northern Canada, especially when urea phosphate is applied. Following the application of urea phosphate, an increase in the number of bacteria was found. This observation shows that microbial activity has been stimulated, which in turn has accelerated the decomposition of various oil components. The recovered oil's continuous decrease in saturated component weight and the quick disappearance of isoprenoids and n-alkanes suggest that the fertilizer-stimulated microbial population is efficiently degrading these hydrocarbon compounds. The breakdown of isoprenoids and n-alkanes, which are frequent components of crude oil, is a crucial part of oil cleanup initiatives [217]. Fertilizer treatment, especially with a carbon-to-nitrogen ratio of 61:1 and a carbon-to-phosphorus ratio of 607:1, produced encouraging effects after just one year, according to the test findings done in Antarctic mineral soils. Comparing the fertilized plots to the untreated ones, a significant rise in microbial activity was observed. The significant degree of microbial activity seen in the fertilized soils indicates that microbial growth and metabolic activities were promoted by the supply of nutrients through fertilization. Consequently, this probably made it easier for organic matter to break down and degrade, including any possible pollutants in the soil [148].

Increased bioremediation activity was seen in a recent experiment carried out at a diesel fuel-contaminated location in Prudhoe Bay, Northern Alaska, where elevated temperatures between 0.5 and 7.8 °C were introduced into the upper subsurface. Raising the temperature probably had a major impact on the microbial activity that broke down the pollutants in diesel fuel. Increased biodegradation rates might result from faster metabolic processes in microbial populations at warmer temperatures. In colder climates like Northern Alaska, where lower temperatures might restrict microbial activity and bioremediation efficacy, this phenomenon is especially noteworthy. Alaska's contaminated media cleanup initiatives included a variety of techniques, such as land farming, microbiological supplementation, fertilizer additions, and constructed systems. The intricacy of remediation difficulties and the requirement for a variety of approaches to successfully address them are reflected in this multifaceted approach [86]. Table 6 lists the applications of bioremediation in various nations worldwide.

Table 6
Bioremediation techniques are implemented in several countries [2].

Country	Biological systems used
USA	Pure or mixed cultures of <i>Bacillus</i> , <i>Pseudomonas</i> , <i>Clostridium</i> , and Gram-negative rods; mixed cultures of hydrocarbon; mixed cultures of marine source bacteria; spore suspension of <i>Clostridium</i> ; slime-forming bacteria; indigenous stratal microflora; ultra micro bacteria.
Venezuela	Modified cultures of mixed enrichment.
Russia	Pure cultures of <i>C. tyrobutiricum</i> ; indigenous microflora of water; injection and water formation bacteria mixed cultures; naturally occurring microbiota of industrial (food) wastes; activated sludge bacteria.
Netherlands	<i>Betacoccus dextranicus</i> (Slime-forming bacteria).
China	Mixed enriched bacterial cultures of <i>bacillus</i> , <i>Eubacterium</i> , <i>Bacteroides</i> , <i>Fusobacterium</i> , <i>Pseudomonas</i> ; slime-forming bacteria: <i>Brevibacterium viscogenes</i> , <i>Xanthomonas campestris</i> , <i>Corynebacterium gumiform</i> .
Saudi Arabia	Sufficient bacterial inoculum based on each technology's needs.
Norway	Naturally existing nitrate-reducing microorganisms in North Sea water.
Bulgaria	Native microorganisms that oxidize oil from water input and creation.
Oman	Autochthonous spore-forming bacteria are found in soil polluted by oil and oil wells.
Former Czechoslovakia	sulfate-reducing bacteria; Hydrocarbon oxidizing bacteria (predominantly <i>Pseudomonas</i> sp.).

11. Implication for future remediation efforts

These revolutionary bioremediation techniques provide practical, eco-friendly ways to lessen the effects of oil spills on ecosystems. It is important to take into account the possible hazards and legal ramifications linked to the utilization of these technologies. Here are a few innovative approaches for oil spill bioremediation:

11.1. Genetically modified microorganisms

The first genetically engineered microorganism (GEM) created in 1970 was indeed a significant milestone in bioremediation research. This engineered microorganism, often referred to as the "superbug," was designed to degrade various toxic hydrocarbons, including hexane, octane, toluene, xylene, camphor, and naphthalene. Scientists were able to improve the microorganism's capacity to digest and break down the pollutants more effectively by adding these genetic alterations [131]. Well-known bacterial strains *Pseudomonas aeruginosa* (NRRL B-5472) and *Pseudomonas putida* (NRRL B-5473) have been the subject of in-depth research due to their capacity to break down a wide range of hydrocarbons. Genes from these strains encode enzymes that break down substances including paraxylene, salicylate, camphor, toluene, methyl toluene, and naphthalene. These strains have two operons, *xyIUWCMABN* and *xyIXYZLTEGFJQKIHSR*, which are essential for the metabolism of toluene, methyl toluene, and paraxylene. These operons encode genes that catalyse the hydrocarbons' breakdown into simpler chemicals that the bacterium may use as a source of carbon and energy. One of the main obstacles to bioremediation is the sluggish pace of degradation of natural microorganisms and their limited capacity to digest some complex organic contaminants. Although native microorganisms may break down pollutants to some extent, their rate of degradation is frequently too sluggish to allow for the efficient remediation of polluted locations in a reasonable amount of time. Microorganisms might not have the exact enzymes or metabolic pathways needed to break down newly introduced pollutants in the environment. Because of this, these pollutants have the potential to linger in the environment for long periods, harming the ecosystem and endangering human health. Chai et al., [52]. Research on designing microbes for particular bioremediation has been aided by recent developments in molecular biology. To enable microorganisms to biodegrade particular contaminants that are not broken down by naturally occurring bacteria adequately or quickly enough, additional genetic traits are added to the microbes via genetic modification. To combine different desired metabolic properties of organisms and manipulate significant genetic factors, microbiological knowledge is required in addition to ecological and biochemical mechanism understanding [128]. Information on the interactions between bacteria and pollutants, their genetic basis, metabolic pathways, operon layout, and molecular biology must all be taken into account for the appropriate design of GEMs [130]. The

process by which the microbial consortia's genetic potential expanded was likewise disclosed by gene sequencing. *Pseudomonas* species are widely distributed in many settings such as soil, water, and sediment. They are well-known for their strong metabolic capacities, which are especially useful in the degradation of harmful substances. *Pseudomonas* bacteria's ability to contain genes for the metabolism of pollutants within their chromosomes and plasmids makes them useful genetic manipulators. Because of this genetic flexibility, scientists have access to a large pool of catabolic genes that they may use for bioremediation.

It is possible to create GEMs with improved biodegradation capabilities, particularly if the underlying genes, enzymes, and degradative processes are well characterized [216]. All things considered, there is a great deal of promise for reducing environmental pollution and boosting the effectiveness of bioremediation procedures through the development of GEMs with improved biodegradation capabilities.

11.2. Combined techniques for oil spill bioremediation

By integrating biological and electrochemical methods, bio-electrochemical systems (BES) enable the bioremediation of pollutants, including oil spills, using the action of specialized bacteria referred to as exoelectrogens, electricigens, or anode-respiring bacteria. These bacteria are special because they can oxidize organic substrates anaerobically while transferring electrons to an electrode (anode) ([32]; L. [144]). This method allows for real-time deterioration process monitoring and is controlled. When it comes to this process, our understanding of the microbial process is less than that of the electron transport mechanism. Another drawback for in-the-field applications is the impact environmental factors can have on the metabolic processes of microorganisms that break down pollutants [146,147,67]. Understanding the microbial mechanism facilitates our comprehension of the two concurrent bioremediation processes: exoelectrogen microbial communities that utilize the electrodes and the natural attenuation process using native electron acceptors in the environment (L. [144]). Benthic microbial BES offers a viable substitute for the traditional remediation procedure in the bioremediation of submerged sediments, as the absence of an electron acceptor poses a significant challenge (H. [137]). However, this technique may also be used successfully for biological remediation of spilled oil in both soil and water[32]. This method was initially used to clean wastewater and then refined to remove difficult-to-remove substances like PH [3].

A more straightforward method known as "Oil-Spill Snorkel" is a unique BES technique utilized for the bioremediation of soil and sediments polluted with PH. The snorkel, an electrode consisting of a conducting substance, is positioned to establish the electrochemical connection in this system. The electrode, or snorkel, is an electrical rod that bridges the gap between the aerobic and anaerobic zones. It functions as both the anode and the cathode. Nevertheless, this approach does not allow for the monitoring or harvesting of electricity. An

anode plate buried in sediments accepts the electrons produced by the oxidation of pollutants. The resulting electrons are moved to the cathode, which has aerobic conditions, via a snorkel. Water is formed when oxygen is reduced [65,67]. Overall, by using exo-electrogens' metabolic activity and fusing it with electrochemical processes, BES offers a potential strategy for the bioremediation of oil spills and other environmental toxins. With this technique, environmental remediation might be accomplished in a sustainable and energy-efficient manner.

11.3. Modern material introduction

In novel approaches biowastes, inorganic materials, polymeric materials, etc. are added to enhance the bioremediation. Biosurfactant addition is another material that recently gained attention in the studies on bioremediation of oil spills [165]. Yeasts, bacteria, and filamentous fungi are among the microorganisms that may create biosurfactants either extracellularly or as a component of the cell membrane. By encouraging the interaction of microorganisms with hydrophobic phases, such as oil droplets or soil particles, these microbial surfactants are essential in augmenting the biodegradation of hydrophobic pollutants. Microbial cells emit extracellular biosurfactants, such as trehalose lipids produced by *Rhodococcus species*, into the surrounding environment [124]. These biosurfactants aid in the emulsification and breakdown of hydrophobic substances by oil-degrading bacteria by decreasing the surface tension of water and increasing their solubility. There is a large range of structural variation in biosurfactants, such as polymeric biosurfactants, phospholipids, neutral lipids, fatty acids, lipoproteins, and glycolipids [28]. The growing consciousness of environmental sustainability has led to a growing interest in utilizing biosurfactants' potential to lessen dependence on conventional chemical-based goods and encourage more sustainable activities [39]. Research has recently focused on the use of biosurfactant microorganisms because they are more ecologically friendly and offer better biodegradability than conventional surfactants [197]. Given that surface-active molecules increase solubility, decrease interfacial tension, and boost microbial activity, they are essential for the removal and biodegradation of hydrophobic chemicals in soil. Because of this, they are useful instruments in environmental rehabilitation projects that try to return damaged areas to their original state [39]. Lipopeptides, which have a hydrophobic fatty acid moiety and a hydrophilic peptide moiety, are amphiphilic biosurfactants and are thus among the most prevalent forms of biosurfactants. Lipopeptides have strong surface-active characteristics, such as lowering surface tension, emulsification, and foaming, because of their special structure. Many genera of bacteria, including *Mycobacterium*, *Acinetobacter*, *Pseudomonas*, and *Bacillus*, are well recognized for producing lipopeptide biosurfactants. The potential uses of these bacteria in several sectors and their ability to produce biosurfactants have been the subject of substantial research [39]. Furthermore, immobilizing microorganisms on a carrier material is a useful way to speed up biodegradation during bioremediation procedures. The process of immobilization entails the physical or chemical attachment of microbes to a solid support material, such as sorbent material or carrier matrix, to enhance their activity, stability, and retention within the polluted environment. Alessandrello et al., [12], the value of organic nutrition sources, such as plant and fish tissue, in promoting the bioremediation of hydrocarbon-contaminated sandy beach sediments. The findings show that when organic nutrients especially fish tissue were added, the rate at which pollutants degraded rose considerably in comparison to when inorganic nutrients were added [108].

By adding organic nutrients to bioremediation techniques, polluted areas can be cleaned up more successfully by increasing microbial activity and hastening the natural attenuation of pollutants.

12. Challenges and limitations

Oil spill response optimization challenges are intricate and multi-dimensional, incorporating variables like the spill's magnitude and location, the surrounding environment, the resources at hand, and the

requirement to reduce threats to human health and ecological integrity.

- There are very few studies that only concentrate on optimization in oil spill response, they focus on creating optimization tactics to enhance response efforts as the significance of good spill response becomes more obvious [177,47].
- Decision-making issues in oil response planning and on-site operations, although the majority concentrate on simulating the weathering and transit of oil [226]. The integration of oil characteristics, the weathering model, and the forecasting model is a topic of limited current literature. Additionally, mixed-integer linear programming model which is crucial for response operations decision-making [163,96]
- The rate of biodegradation is slowed down by cold temperatures, which also slow down microbial activity. Restoration process are the physical barriers caused by snow, ice, and permafrost, which make it difficult for nutrients and microbes to enter the soil. In cold climates, snow, ice, and frozen ground can serve as physical barriers to prevent oil spills from spreading and penetrating [155,41]. Because of the obstruction caused by the snow and ice that is already there, oil can be effectively sequestered over the winter months with minimal weathering, favouring manual cleaning techniques. A comparison test, however, revealed that the presence of ice maintained the hydrocarbon degraders at a level around ten times lower than in systems without ice [155]. Nutrient shortages, and frozen soil can seriously impair the efficacy of bioremediation.
- Melting rain and snow can spread oil horizontally and longitudinally; this released oil will stay trapped and might eventually become an ongoing source of oil slicks in the runoff waters. Because of the nonlinear connection of temperature and water systems, mass transfer, and metabolic activities, freeze-thaw events hamper bioremediation.
- When it comes to detecting oil spills, there are several challenges. Water, salt, air, and silt, all of which might give misleading signals to the detecting devices about the presence of oil in the ice [100].
- Oil spill monitoring in hostile conditions faces substantial problems due to the various nature of ice kinds and their crystalline orientations.
- During offshore spill response operations that depend on visual tracking, planning, and surveillance, visibility is essential. The efficiency of these activities can be seriously hampered by poor visibility brought on by fog, rain, snow, or darkness. This makes it challenging for response teams to determine the exact amount of the spill, follow its progress, and put in place the proper containment and cleanup procedures [105].
- The majority of the contamination of groundwater is caused by leaks from underground oil storage tanks [116]. There are now several methods available for cleaning up PH pollution in various environmental matrixes. Groundwater PH pollution was formerly treated ex-situ using the pump and treat method; however, this approach was less successful in terms of economy and practicality. In groundwater aquifers, the destiny of PH is more complex and impacted by a range of biotic and abiotic activities.
- Phytoremediation has the potential to be a cost-effective and sustainable approach to environmental cleanup, its success depends on careful planning, site-specific considerations, and ongoing monitoring and management [14,152,93]. Numerous elements, including the kind, extent, and depth of the polluted site, the contaminated medium, the site's environment, the kind of plants, and agronomic techniques, influence the overall cost of a phytoremediation system [204]. PH in various locations show a variety of mixes, ranging from highly mobile extremely hydrophobic [122]. Consequently, a wide variety of phytoremediation costs can be incurred. Entire cost of phytoremediation comprises the following: design, setup, yearly operation, monitoring, and management [83]. One of the limitations of phytoremediation is that it typically

requires a longer duration to achieve cleanup standards compared to some alternative technologies [204,83].

- There are benefits and drawbacks to each kind of oil-sorbent material in terms of availability, price, and performance quality [122]. The non-biodegradability of synthetic sorbents can provide a challenge to the sustainability of the environment, especially when it comes to disposing of spent sorbent materials after they have been used to absorb pollutants [140]. Conversely, research has demonstrated that inorganic sorbents are not very effective in holding onto oil and keeping it from leaking into the environment. Furthermore, inorganic sorbent minerals may have detrimental consequences on seabed ecosystems [51]. Managing big sorbents can indeed provide serious difficulties that could compromise the effectiveness of operations for spill response and cleanup. Large sorbents can be heavy and challenging to move, especially in small spaces or places that are difficult to reach, like rocky shorelines, marshes, or wetlands [121,59].

It will need multidisciplinary cooperation between scientists, engineers, and stakeholders to address these issues and create cutting-edge technology and monitoring strategies suited to the particularities of ice settings [87,110]. Because of the differences in their physical and chemical characteristics, crude oils and petroleum-based goods behave quite differently when spilled in cold and severe settings. These characteristics have an impact on the methods and tools chosen for monitoring and sampling [102]. Toxicological tests using common test organisms are often the foundation for evaluating the environmental risk associated with contaminants [206]. Hence, comprehending the idea of PH's bioavailability will help with the development of a successful risk assessment plan, which in turn will give important information for creating and planning remediation strategies for PH-contaminated locations. How well a remediation strategy works is determined by how bioavailable the contaminants are.

13. Conclusion and future aspects

Every nation's economic growth was significantly influenced by the oil trade. The regulation of the global oil trade and the damage caused by oil spills from ships need constant reinforcement. On the other hand, oil spills have severely harmed the marine natural ecosystem in addition to causing enormous financial losses and waste of resources. One increasingly common method for clearing polluted soil and aquifers is bioremediation. Scientists are looking at creative engineering approaches to deal with these problems. For instance, the advancement of methods such as gas sparging has improved petroleum product aerobic degradation. It is anticipated that improvements in bioaugmentation technology would enhance cell genetic manipulation, beyond microbiological constraints. These limitations include handling and stimulating bacteria with difficulty, ensuring appropriate contact between microbes and pollutants, and ambiguities surrounding the behaviour of microbes in the field. By enhancing the interaction between bacteria and pollutants, techniques including heat injection solubilization of waste, intense-pressure subsurface matrix fracturing, and surfactant addition are intended to improve the effectiveness of bioremediation. Developing adapted techniques based on molecular biology expertise for quick and economical evaluation is one of the future research objectives. The physiochemical parameters in situ characterization show potential for transforming field assessment testing. Because it offers financial advantages and a lesser danger of pollutant spread, bioremediation is a commonly recognized approach for environmental cleaning. However, some people are reluctant to choose bioremediation as their preferred method of waste management due to the need for careful handling and ongoing monitoring of bacteria in underground habitats. In general, the development of quick and effective methods to address present issues and move toward a cleaner environment is what will determine the direction of bioremediation in the future. In summary, although there are obstacles to overcome, natural oil spill repair is a viable and eco-friendly method of reducing environmental

harm. We can better plan for and respond to oil spill situations in the future and protect our priceless ecosystems and natural resources for future generations by tackling these issues and adopting creative solutions.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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